

A Unified Mathematical Model for Truth-Probability and Importance-Scoring of Statements

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Abstract

This paper presents a unified mathematical framework for assigning two scores to any statement s made by a source to a receiver r : a truth-probability score and an importance score. The framework produces three outputs — $T_abs(s)$, the receiver-independent evidential truth posterior (the paper’s shorthand: objective truth-probability); $T_perc(s,r)$, the perceived truth-probability as modulated by delivery, context, and receiver properties; and $I(s,r)$, the importance score, which is always receiver-dependent. The central architectural claim is that the input space of T_abs is a proper subset of the input space of T_perc , which is a proper subset of the input space of I , forming a unified scoring space where the truth scores are modeled, conjecturally, as projections of the importance tensor onto specific axes. All inter-cluster updates operate via sequential Bayesian updating with a likelihood ratio function derived from cluster-specific neutral points. The framework integrates six structured input clusters covering source credibility, statement properties, delivery context, situational context, receiver properties, and corroboration. Formal axioms ground every architectural decision. Open problems are identified and precisely stated.

Methodological status. This is a formal architecture and research program, not an empirically calibrated scoring instrument. Its numerical constants are exposed as defaults pending calibration, not hidden as validated parameters.

1. Introduction

The problem of assigning formal truth-probabilities to statements has been approached from multiple directions in the literature: Bayesian epistemology treats belief as subjective probability updated on evidence; formal pragmatics examines how context modulates meaning; source credibility research identifies trust and expertise as primary credibility dimensions. What does not exist is a single unified framework that formally separates objective from perceived truth-probability, treats both as projection-like restrictions of a richer importance space, and provides a computationally tractable scoring architecture grounded in information theory and Bayesian inference.

This framework attempts to fill that gap. It is both normative — specifying how truth and importance should be weighted given rational epistemic standards — and empirically tractable, drawing on observable inputs organized into six structured clusters.

The framework makes three contributions. First, it introduces a principled three-output architecture (T_abs , T_perc , I) with a formally established input-space hierarchy. Second, it derives likelihood ratio functions for Bayesian updating from cluster-specific neutral points, eliminating free parameters. Third, it identifies and patches a topological hole in naive falsifiability-consistency formulations that would allow perfectly structured but empirically inert claims to achieve high truth-probability.

1.1 Related Work and Positioning

This paper sits at the intersection of Bayesian epistemology, formal social epistemology, testimony, pragmatics, credibility theory, and decision-theoretic relevance. Bayesian epistemology supplies the basic idea that rational belief can be modeled as probability and updated by evidence, from the classical formulation of Bayes’s rule to subjective and decision-theoretic accounts of credence (Bayes, 1763; Ramsey, 1931; de Finetti, 1937; Jeffrey, 1983; Joyce, 1998; Bovens and Hartmann, 2003). The present framework inherits that probabilistic discipline but does not reduce everything to one posterior. It explicitly separates the receiver-independent evidential posterior, the posterior perceived by a receiver, and the importance of the statement for that receiver.

Social epistemology and the epistemology of testimony supply the second foundation. Work on testimony and source credibility emphasizes that hearers do not receive bare propositions; they receive propositions from sources with histories, expertise, incentives, social positions, and relations to the hearer (Coady, 1992; Goldman, 1999; Fricker, 1994; Lackey, 2008; Sperber et al., 2010). This framework preserves that insight in Cluster 1 and Cluster 5: source credibility is statement-specific rather than source-universal, and receiver response is explicitly receiver-relative rather than treated as a universal distortion term.

Formal pragmatics and relevance theory supply the third foundation. Statements are not only true or false; they are uttered in contexts, under constraints, with different degrees of relevance to different receivers (Grice, 1975; Sperber and Wilson, 1995). The delivery and situational clusters are designed to formalize that pragmatic layer without letting it contaminate receiver-independent truth evidence. Delivery and context can move perceived truth and importance; they do not directly change the receiver-independent truth posterior.

The credibility literature supplies the fourth foundation. Classic communication research treats expertise and trustworthiness as central credibility dimensions (Hovland and Weiss, 1951; Hovland et al., 1953), and later work on digital credibility makes explicit that interface, medium, and presentation cues affect belief even when the underlying claim is unchanged (Fogg and Tseng, 1999). This framework turns those ideas into separable clusters rather than leaving them as informal modifiers.

Finally, the framework borrows from information theory and risk/decision analysis. KL-divergence is used as a formal novelty measure, while the importance score reflects consequence, stakes radius, and receiver impact rather than truth alone (Shannon, 1948; Kullback and Leibler, 1951; Kaplan and Garrick, 1981). The novelty here is not that any one component is unprecedented. The claim is architectural: the paper places truth-probability, perceived truth-probability, and importance in one nested scoring space while forcing each to use only the variables it is entitled to use.

2. Philosophical Foundation and Core Architecture

2.1 The Three-Output Architecture

The framework produces three scores for any statement s and receiver r :

- $T_abs(s)$ in $[0,1]$: receiver-independent evidential truth posterior, called objective truth-probability as shorthand.
- $T_perc(s,r)$ in $[0,1]$: perceived truth-probability, receiver-dependent.
- $I(s,r)$ in $[0,1]$: importance score, always receiver-dependent.

The central structural claim is the strict input-space hierarchy. Let I_abs , I_perc , and I_I denote the input spaces of T_abs , T_perc , and I respectively:

$$I_{abs} \subset I_{perc} \subset I_I.$$

T_abs , T_perc , and I are real-valued scoring functions defined on these nested input spaces. Every input that determines T_abs also determines T_perc and I , and every input that determines T_perc also determines I , but not conversely. T_abs and T_perc are modeled as projection-like restrictions of I onto specific axes of

the unified scoring space — obtained, conjecturally and pending the full projection proof, by restricting the domain of I to I_abs and I_perc respectively.

2.1.1 Conjectural Status of the Projection Claim

The strict input-space hierarchy is part of the formal architecture and is established by construction. The stronger claim that T_abs and T_perc are literally recoverable as projections of the full importance space is more cautious: it is the guiding projection interpretation of the model, not yet a completed theorem. For that reason, the paper treats the projection language as a conjectural structural interpretation. The framework motivates and models truth scores as projections or restrictions of the richer importance space, while the full proof that these scalar truth scores are obtainable as projections of I is kept as an explicit open problem. This prevents the central architecture from pretending to have already proved more than it has.

2.2 Formal Axiom System

The following axioms ground every architectural decision in the framework.

Axiom 1 (Receiver-Independent Evidential Truth Posterior): For any statement s , there exists a fixed probability $T_abs(s)$ in $[0,1]$ representing the receiver-independent evidential posterior for the claim’s truth, not truth itself. It is independent of any receiver’s psychology, delivery conditions, or personal stakes.

Axiom 2 (Receiver Dependence of Perception): For any statement s , perceived truth $T_perc(s,r)$ is a function of both s and receiver r . No two receivers are guaranteed to produce identical T_perc for the same s .

Axiom 3 (Receiver-Independent Truth as Perception Limit): As receiver knowledge P approaches 1 and all perceptual distortions vanish, $T_perc(s,r)$ approaches $T_abs(s)$.

Axiom 4 (Importance Undefined Without Receiver): For any statement s , $I(s)$ without a specified receiver r is undefined. Importance is strictly a function $I(s,r)$.

Axiom 5 (Strict Superset Hierarchy): Every input that determines $T_abs(s)$ also determines $T_perc(s,r)$ and $I(s,r)$. Every input that determines $T_perc(s,r)$ also determines $I(s,r)$. The converse does not hold in either case.

Axiom 6 (Evidence Monotonicity): For any statement s , additional independent corroborating evidence never decreases $T_abs(s)$. Additional contradicting evidence never increases $T_abs(s)$.

Axiom 7 (Bias as Self-Amplifying Corruptor): A source with maximum bias (B_eff approaching 0) has zero credibility regardless of all other source properties. Bias is not a linear penalty on credibility but a self-amplifying corruptor: the credibility damage of bias scales with the degree of bias itself.

Axiom 8 (Falsifiability as Truth Condition): A statement that is in principle unfalsifiable cannot be assigned a non-zero objective truth-probability by this framework. Furthermore, unfalsifiability breaks the Bayesian update engine: if a claim cannot be tested, the likelihood ratio of any evidence observation equals 1, and the posterior never updates from the prior.

Axiom 9 (Self-Report Inferiority): Self-reported signals carry strictly less evidential weight than structurally verifiable signals of the same magnitude.

Axiom 10 (Expertise as Prerequisite): A source with zero domain expertise has zero credibility on statements in that domain regardless of all other source properties.

Axiom 11 (Importance Humanity Principle): Perceptual and receiver factors contribute strictly more to importance than to perceived truth. Receiver-independent truth evidence contributes strictly less to importance than to perceived truth.

Axiom 12 (Structural Signal Dominance Ordering): Among delivery signals, linguistic register carries strictly more evidential weight than tone, which carries strictly more than expression.

Axiom 13 (Stakes Radius Principle): Among situational signals, the evidential weight of stakes is proportional to the radius of affected parties. Broad societal stakes outweigh speaker stakes, which outweigh background situation.

Axiom 14 (Opposition Superlinearity): The presence of active opposition is a superlinear negative signal on perceived truth. Its evidential weight grows faster than linearly with opposition strength.

3. Variable Master List

All inputs are normalized to $[0,1]$ unless stated otherwise. Higher values always represent the more credibility-positive or truth-positive direction.

3.1 Cluster 1 — Source Credibility

Symbol	Variable	Description
$B = (B1, B2, B3, B4)$	Bias vector	Domain-specific bias: B1=political, B2=financial, B3=institutional, B4=motivated reasoning. 0=fully biased, 1=no bias.
$B_eff(s)$	Effective bias scalar	Statement-weighted collapse: $B_eff(s) = \text{sum}(w(s,i)*Bi) /$ $\text{sum}(w(s,i))$
$C = (C1, C2, C3, C4)$	Expertise vector	C1=formal credentials, C2=peer recognition, C3=years practice, C4=novel contributions track record.
$C_eff(s)$	Effective expertise scalar	Same weight vector as B_eff : $C_eff(s) = \text{sum}(w(s,i)*Ci) /$ $\text{sum}(w(s,i))$
D	Rank relative to receiver	Social, institutional, epistemic standing of source relative to this specific receiver.
E	Stated confidence	Self-reported certainty. Carries less weight than D by Axiom 9.
$w(s,i)$	Domain relevance weights	Derived from primary and adjacent subjects of statement s (Cluster 2). Same vector serves both B_eff and C_eff .

3.2 Cluster 2 — Statement Properties

Symbol	Variable	Description
F	Novelty-penalty field-fit score	Truth-positive transform of novelty relative to current knowledge in its domain. Higher F means the claim is less penalized by field-level surprise; raw surprise can be written as $F_raw = 1-F$. Used in T_abs only.

G	Internal logical consistency	Degree to which statement contains no internal contradictions. Necessary but insufficient for truth.
H	Falsifiability	Degree to which statement is in principle testable and disprovable. Tethers claim to empirical reality.
J	Specificity	Degree to which statement makes a precise, bounded claim.
k	Falsifiability steepness	$k = \ln(100) / 1 = 4.605$, derived so that max of $(1 - e^{(-kH)}) = 0.99$ at $H=1$.

3.3 Cluster 3 — Delivery Context

Symbol	Variable	Description
S	Language register	Match between linguistic register and receiver's comprehension and epistemic expectations.
T	Tone	Appropriateness of affective tone to content and context.
U	Posture and expression	Congruence of nonverbal signals with statement. Weakest delivery signal.
V	Time context	Receptivity of temporal setting — time of day, week, month, year.
W	Location	Institutional and physical weight of setting. A White House briefing and a street corner carry categorically different scores.
X	Public vs private setting	Degree to which statement is made in a publicly observable, institutionally verifiable context. $X=0$ means statement to oneself: $Deliv=0$ by limit.

3.4 Cluster 4 — Situational Context

Symbol	Variable	Description
Y	General situation	Background charge of situational context around the statement's subject at time of delivery.
Z	Speaker stakes	What speaker risks by making this statement. Double-edged: handled via Z_{eff} .
Z_{eff}	Bias-adjusted stakes	$Z_{eff} = Z * (1 - Z * B_{eff}(s))$. High stakes from biased source discounted; high stakes from clean source amplifies credibility.
AA	Audience composition	Receptivity, expertise, and alignment of present audience.

AB	Opposing voices	Presence of active counterarguments. 0 = maximum opposition, 1 = no opposition.
AD	Broad stakes	Stakes for all parties beyond speaker: audience, institutions, society.

3.5 Cluster 5 — Receiver Properties

Symbol	Variable	Description
N	Novelty relative to receiver	$N = KL(P_s Q_r)$. P_s = distribution implied by statement, Q_r = receiver's current belief distribution. Enters T_{perc} as N_{perc} ; enters I as N_{imp} .
N_{perc}	Novelty for T_{perc}	$N_{perc} = 4.18 * N_{norm}^{0.7} * (1 - N_{norm})^{1.5}$, where $N_{norm} = 1 - e^{(-N)}$. Inverted U: peaks at ~30% of normalized scale.
N_{imp}	Novelty for $I(s,r)$	$N_{imp} = 1 - e^{(-N(1+N))}$. Monotonically increasing — more novelty always improves importance.
P	Prior knowledge	Receiver's existing domain knowledge. High P pulls T_{perc} toward T_{abs} .
Q_{rel}	Relationship to speaker	Trust, authority, emotional bond, social proximity between receiver and source.
R	Cognitive/emotional state	Degree of emotional engagement. Follows inverted U — moderate engagement optimal, extremes distort.
AC	Personal consequence	Degree to which statement's truth would materially change receiver's world. $AC=0$ collapses Recv to 0.

3.6 Cluster 6 — Corroboration

Symbol	Variable	Description
$K_1...K_n$	Independent sources	n independent corroborating sources. Each contributes one Bayesian update with diminishing likelihood ratios.
L	Established knowledge consistency	Degree to which statement coheres with verified existing knowledge in its domain. $x0_L = 0.75$.
M	Physical/empirical evidence	Presence of direct empirical evidence. High likelihood ratio — hard to fabricate. $x0_M = 0.50$, floor 0.40 context-dependent.

4. The Bayesian Update Engine

4.1 Prior — Rolling Average

The prior $P(A)$ for any statement s by a source is the rolling moving average of that source's last $w=10$ evaluated statements:

$$P(A) = (1/w) * \sum_{i=1}^w T_{\text{abs}}(s_i)$$

Cold-start anchor: $P(A) = 0.50$ for first $w-1$ statements, or a domain-specific baseline where available. This encodes historical accuracy as mathematical momentum without the double-counting paradox that would result from including past accuracy as both prior and update evidence.

4.2 Likelihood Ratio Function

All cluster scores x in $[0,1]$ translate to likelihood ratios via:

$$\text{Lambda_c}(x) = \exp((x - x0_c) / (x0_c * (1 - x0_c)))$$

Where $x0_c$ is the cluster-specific neutral point — the score at which the prior is unchanged. This function satisfies:

- $x = x0_c$
to $\text{Lambda} = 1$
to prior unchanged
- $x > x0_c$
to $\text{Lambda} > 1$
to evidence supports A
- $x < x0_c$
to $\text{Lambda} < 1$
to evidence against A
- Bounded at all x in $[0,1]$ — no singularities
- No free constants — fully determined by $x0_c$ alone

Posterior odds after each update:

$$\text{Odds}(A|x) = \text{Lambda_c}(x) * \text{Odds}(A)$$

Recovered to probability via $P = \text{Odds} / (1 + \text{Odds})$.

4.3 Diminishing Corroboration

To prevent echo chambers and coordinated source inflation, the likelihood ratio of the n -th independent source decays asymptotically:

$$\text{Lambda_n} = 1 + (\text{Lambda_base} - 1) / n^{\alpha}$$

Where α controls decay speed. Currently undefined — Open Problem 5. The 20th corroborating source moves the posterior substantially less than the 1st.

5. Cluster Scoring Functions

5.1 Cluster 1 — Source Credibility: $\text{Cred}(s)$

$$\text{Cred}(s) = B_{\text{eff}}^{(B_{\text{eff}}^{-0.5})} * \sqrt{C_{\text{eff}}(s)} * (4D + E) / 5$$

Structural encoding:

- $B_eff^{(B_eff^{(-0.5)})}$: bias self-amplifies its own penalty. The exponent $B_eff^{(-0.5)} = 1/\sqrt{B_eff}$ grows as B_eff shrinks — a more biased source is punished harder by its own bias. B_eff approaching 0 collapses Cred to 0 by limit definition (Axiom 7).
- $\sqrt{C_eff}$: expertise enters concavely. Zero to some expertise is a large credibility jump; high to very high is marginal. $C_eff = 0$ collapses Cred to 0 (Axiom 10).
- $(4D+E)/5$: rank dominates stated confidence 4:1 by Axiom 9. Normalization: $4+1=5$. Note: past accuracy A removed from this formula and fully encoded in the rolling prior, eliminating the Bayesian double-counting paradox.

Boundary verification: All inputs = 1 gives Cred = 1. B_eff approaching 0 gives Cred approaching 0 by limit. $C_eff = 0$ gives Cred = 0.

5.2 Cluster 2 — Statement Properties: Stmt(s)

$$\text{Stmt}(s) = \left(\frac{F^{1.4} + J}{2} \right) G(1 - e^{-kH}).$$

Where $k = \ln(100) / 1 = 4.605$, derived so that the falsifiability term reaches 0.99 at $H=1$. Here F is the truth-positive novelty transform, not raw surprise: raw surprise lowers F , and therefore lowers Stmt.

Structural encoding:

- $F^{1.4}$: novelty penalized through the truth-positive transform F — the model is epistemically cautious about extraordinary claims. Since raw surprise lowers F , the exponent 1.4 makes strong field-level surprise costly rather than rewarding it. Extraordinary claims do not get rewarded easily.
- J : specificity enters linearly — taken more at face value than the novelty-penalty transform.
- G : consistency as independent multiplier, separated from falsifiability. Internal coherence is necessary but not sufficient for truth.
- $(1 - e^{-kH})$: falsifiability tether. As H approaches 0, this term approaches 0, collapsing Stmt regardless of how consistent G is. This closes the topological hole in the original formulation where $G=1$ and $H=0$ would give $\text{Stmt}=1$, allowing perfectly consistent but unfalsifiable claims (simulation theory, unfalsifiable conspiracy theories) to score at maximum. The separation of G and H into independent multipliers reflects that they measure categorically different things: internal coherence versus empirical testability.

Remark: the original formulation $G^{(1/\sqrt{H})}$ coupled consistency and falsifiability in one exponent, allowing $G=1$ to override $H=0$. The corrected form prevents this by making falsifiability an independent multiplicative gate.

5.3 Cluster 3 — Delivery Context: Deliv(s,r)

$$\text{Deliv}(s,r) = W * (V * X^{1.1})^{1.1} * ((3S + 2T + U) / 6)^{0.5}$$

Structural encoding:

- $W * (V * X^{1.1})^{1.1}$: location is the base, modulated jointly by time context and public/private setting. $X^{1.1}$ slightly super-weights setting over time. Institutional context dominates delivery.
- X approaching 0: Deliv approaches 0 by limit. A statement made to oneself is not communicative transmission and has no delivery score.
- $(3S+2T+U)/6$: language register dominates tone dominates expression, 3:2:1 by Axiom 12. Normalization: $3+2+1=6$. Language carries the most structural information about framing and epistemic register.

- $\wedge 0.5$: congruence signals enter with square-root concavity — present and meaningful but not dominant over contextual signals.

5.4 Cluster 4 — Situational Context: $\text{Sit}(s,r)$

$$\text{Sit}(s,r) = \text{AB}^{\wedge 2.1} * \text{AA}^{\wedge 0.2} * (2 * \text{Z_eff} + 3 * \text{AD} + \text{Y}) / 6$$

Where: $\text{Z_eff} = \text{Z} * (1 - \text{Z} * \text{B_eff}(s))$

Structural encoding:

- $\text{AB}^{\wedge 2.1}$: opposing voices veto superlinearly by Axiom 14. $\text{AB}=0.75$ gives 0.530 — even moderate opposition significantly penalizes perceived truth.
- $\text{AA}^{\wedge 0.2}$: audience composition dampens very weakly — present but not dominant.
- $(2 * \text{Z_eff} + 3 * \text{AD} + \text{Y})/6$: broad stakes dominate (3), speaker stakes second (2), general situation background (1) by Axiom 13. Normalization: $2+3+1=6$. Importance radiates outward — widest circle dominates.
- Z_eff cross-cluster dependency: high stakes from a biased source is discounted. High stakes from a clean source amplifies credibility. The interaction $\text{Z} * (1 - \text{Z} * \text{B_eff})$ captures the double-edged nature of speaker stakes.

5.5 Cluster 5 — Receiver Properties: $\text{Recv}(s,r)$

$$\text{Recv}(s,r) = [\text{N_perc}^{\wedge (1-P)} * \text{Q_rel} * 4\text{R}(1-\text{R})] * \text{AC}^{\wedge 1.15}$$

Structural encoding:

- $\text{N_perc}^{\wedge (1-P)}$: novelty impact decays as prior knowledge grows. A fully knowledgeable receiver (P approaching 1) is unaffected by novelty — they have seen it all.
- Q_rel : relationship multiplies directly at face value — trust and emotional bond enter without dampening.
- $4\text{R}(1-\text{R})$: inverted U for cognitive/emotional state. Maximized at $\text{R}=0.5$ giving value 1. Collapses at $\text{R}=0$ (no engagement) and $\text{R}=1$ (complete emotional overwhelm). Optimal reception requires moderate engagement.
- $\text{AC}^{\wedge 1.15}$: personal consequence slightly superlinear. High consequence amplifies meaningfully. $\text{AC}=0$ collapses Recv to 0 — a statement with zero personal consequence to the receiver registers nothing.
- $\text{N_perc} = 4.18 * \text{N_norm}^{\wedge 0.7} * (1 - \text{N_norm})^{\wedge 1.5}$ where $\text{N_norm} = 1 - e^{\wedge (-N)}$. Three-phase behavior: small novelty (0-30% of normalized scale) helps T_perc ; excess novelty (above 30%) hurts as receiver resistance grows; extreme novelty collapses. Peak at $\text{N_norm} = 0.7/(0.7+1.5) = 0.318$, approximately 30% threshold.

6. $\text{T_abs}(s)$ — Objective Truth-Probability

$$\text{T_abs}(s) = \text{P}(\text{A} \mid \text{Cred}(s), \text{Stmt}(s), \text{K1...Kn}, \text{L}, \text{M})$$

A sequential Bayesian posterior updated over all objective evidence clusters.

6.1 Update Sequence

- Prior: $\text{P}(\text{A})$ = rolling average of source's last $w=10$ evaluated statements. Cold-start: 0.50.
- Update 1 — $\text{Cred}(s)$: $\text{Odds}(\text{A}|\text{Cred}) = \text{Lambda}(\text{Cred}) * \text{Odds}(\text{A})$

- Update 2 — Stmt(s): $\text{Odds}(A|\text{Cred}, \text{Stmt}) = \text{Lambda}(\text{Stmt}) * \text{Odds}(A|\text{Cred})$
- Update 3...n+2 — K1...Kn: each independent source applies one update with $\text{Lambda_n} = 1 + (\text{Lambda_base} - 1)/n^\alpha$. Diminishing returns.
- Update n+3 — L: consistency with established knowledge. $x0_L = 0.75$.
- Update n+4 — M: physical evidence. $x0_M = 0.50$ (floor 0.40 context-dependent). High likelihood ratio — physical evidence is hard to fabricate.

6.2 Key Properties

- Fully receiver-independent.
- Path-independent: order of updates is commutative. Only accumulated evidence matters. For the purposes of this sequential engine, we assume conditional independence of the cluster evidence given the hypothesis A. Relaxing this assumption to model cross-cluster covariance is reserved for future work (see Open Problem 3).
- T_abs feeds back into the rolling prior $P(A)$ for future evaluations of the same source — A is a living estimate.
- Same source receives different Cred(s) for different statements via B_eff(s) and C_eff(s) — credibility is statement-specific, not source-universal.
- Cluster 6 is the Bayesian engine, not a scoring formula. Corroboration is what sequential updating was built to handle.

7. T_perc(s,r) — Perceived Truth-Probability

$$T_perc(s,r) = P(A \mid T_abs(s), \text{Deliv}(s,r), \text{Sit}(s,r), \text{Recv}(s,r))$$

7.1 Update Sequence

- Prior: T_abs(s) — objective posterior as anchor.
- Update 1 — Deliv(s,r): $\text{Odds} = \text{Lambda}(\text{Deliv}) * \text{Odds}(T_abs)$
- Update 2 — Sit(s,r): $\text{Odds} = \text{Lambda}(\text{Sit}) * \text{prior odds}$
- Update 3 — Recv(s,r): $\text{Odds} = \text{Lambda}(\text{Recv}) * \text{prior odds}$

7.2 Key Properties

- T_abs is the anchor — perception deforms it but always starts from it.
- T_perc can exceed or fall below T_abs depending on delivery, situation, and receiver state.
- High P in Recv(s,r) compresses novelty toward zero and pulls T_perc back toward T_abs — knowledgeable receivers converge toward objective truth (Axiom 3).
- Novelty relative to receiver (N_perc via KL divergence) lives in Cluster 5, not T_abs. KL divergence formalizes how surprising a statement is to this specific receiver's existing belief distribution.

8. I(s,r) — Importance Score (Revised Architecture)

$$I_{\text{context}}(s, r) = P(\text{important} \mid \widetilde{\text{Deliv}}(s, r), \widetilde{\text{Sit}}(s, r), \widetilde{\text{Recv}}(s, r)), \quad P(\text{important}) = 0.50.$$

$$O_{\text{context}} = \frac{I_{\text{context}}}{1 - I_{\text{context}}}, \quad \Lambda_{I,T}(T_{\text{abs}}) = \exp\left(\beta \frac{T_{\text{abs}}^{0.5} - x_0}{x_0(1 - x_0)}\right), \quad \beta = 0.15, \quad x_0 = 0.5.$$

$$O_I(s, r) = \Lambda_{I,T}(T_{\text{abs}})O_{\text{context}}, \quad I(s, r) = \frac{O_I(s, r)}{1 + O_I(s, r)}.$$

Fundamental revision from original architecture. The prior is now flat $P(\text{important})=0.50$ — importance does not start from truth, it starts from uncertainty about consequence. T_{abs} enters as one update among equals, not as the prior anchor. The update is performed in odds space and converted back to probability, so $I(s, r)$ remains in $[0, 1]$ by construction. This prevents near-zero T_{abs} from collapsing importance to zero for false but consequential statements while also preventing near-unity T_{abs} from pushing importance outside the unit interval. Delivery, situation, and receiver enter via tilde transforms — a bounded nonlinear mapping that preserves zero at zero, reaches one at one, and grows fast near high values.

8.1 Tilde Transform Definition

Term	Alpha	x0_raw (pre-tilde)	x0_tilde (post-tilde, used in Lambda_c)
Deliv_tilde	0.8	0.379	0.372
Sit_tilde	0.75	0.436	0.415
Recv_tilde	0.7	0.450	0.405

8.2 Key Properties

- Always receiver-dependent — importance without a receiver is undefined (Axiom 4).
- Prior is flat $P(\text{important})=0.50$ — importance does not start from truth. A false statement can be enormously important; truth is one signal among equals, not the anchor.
- T_{abs} enters via a dampened Bayesian odds update with dampening factor $\beta=0.15$. The likelihood ratio is: $\Lambda_{I,T}(T_{\text{abs}}) = \exp(\beta(T_{\text{abs}}^{0.5} - x_0)/(x_0(1 - x_0)))$, where $x_0=0.5$ and $\beta=0.15$. Near-zero T_{abs} gives a mild negative update (Lambda approximately 0.75) rather than a catastrophic penalty — false but consequential statements are not crushed. Near-unity T_{abs} gives a mild positive update (Lambda approximately 1.35) rather than an overwhelming boost — boring truths with low context are not inflated. Delivery, situation, and receiver engagement remain the dominant drivers of importance. The dampening factor $\beta=0.15$ formally instantiates Axiom 11: T_{abs} exerts strictly less evidential weight on I than it does on T_{perc} (where it enters as the undampened prior). In the T_{perc} engine T_{abs} carries maximum evidential weight as the anchor; in the I engine it carries reduced weight $\beta=0.15 < 1$ as one update among equals.
- Tilde transform: $\tilde{X}_\alpha = 1 - \exp(1 - (1 - X)^{-\alpha})$. Bounded in $[0, 1]$. $X=0$ gives tilde=0. $X=1$ gives tilde=1. Grows fast near high values — small improvements at high delivery/situation/receiver produce large importance jumps. Alpha values: Deliv alpha=0.8, Sit alpha=0.75, Recv alpha=0.7.
- T_{abs} contributes strictly less to I than to T_{perc} , as required by Axiom 11 and formally instantiated by $\beta=0.15 < 1$ — importance is more human and contextual than truth.
- $N_{\text{imp}} = 1 - e^{-(N(1+N))}$ replaces N_{perc} in the Recv computation for I . N_{imp} is monotonically increasing — more novelty always improves importance, with no penalty for high novelty.

9. Cross-Cluster Dependency Map

The framework is an interdependent system, not a linear pipeline. The following dependencies exist:

From	To	What flows
Cluster 2	Cluster 1	$w(s,i)$ derived from primary and adjacent subjects feeds both $B_eff(s)$ and $C_eff(s)$. Same weight vector serves both collapses.
Cluster 1	Cluster 4	$B_eff(s)$ feeds $Z_eff = Z*(1-Z*B_eff)$. Speaker stakes discounted by source bias.
Cluster 5 internal	Cluster 5	P modulates own cluster output via $N_perc^{(1-P)}$ term.
T_abs	Cluster 1 prior	T_abs feeds back into rolling prior $P(A)$ for future evaluations of same source. A is a living estimate.
Cluster 5	T_perc and I	N splits: N_perc (inverted U) for T_perc; N_imp (monotonic) for I.

10. Neutral Point Table

Every variable has a cluster-specific neutral point $x0_c$ — the score at which the likelihood ratio equals 1 and the prior is unchanged. These are the semantic judgments of what constitutes a neutral observation for each input.

Cluster	Variable	$x0_c$	Semantic justification
1	B_eff	0.80	Bias at 0.5 is already significantly compromised. High bar for neutrality.
1	C_eff	0.60	Some expertise is the baseline expectation for a source making a claim.
1	D	0.50	Rank is genuinely symmetric — equal rank is truly neutral.
1	E	0.50	Moderate confidence is genuinely neutral — neither over nor underconfident.
2	F	0.40	Moderate novelty is already a positive signal; low novelty adds nothing.
2	G	0.85	Consistency is a baseline expectation. Half-consistent statements are deeply problematic.

2	H	0.80	High falsifiability expected. Moderate falsifiability is already epistemically weak.
2	J	0.55	Moderate specificity is acceptable. More symmetric than G or H.
3	S	0.60	Language mismatch is immediately noticeable. Appropriate register is expected.
3	T	0.55	Tone is more forgiving than language. Moderate deviation tolerated.
3	U	0.52	Expression is weakest signal. Nearly symmetric with slight lean toward congruence.
3	V	0.55	Wrong timing is noticeable but not catastrophic.
3	W	0.60	Location carries institutional weight. Appropriate setting is expected.
3	X	0.65	Public setting is expected for credible statements.
4	Y	0.51	General situation is nearly neutral background. Barely leans toward some charge.
4	Z_eff	0.60	Some stakes expected. High stakes from clean source is a positive signal.
4	AA	0.60	Receptive audience is expected in normal communication.
4	AB	0.75	Absence of opposition is expected for credible statements. Even moderate opposition is concerning.
4	AD	0.40	Broad stakes present is a positive signal. Low broad stakes is genuinely neutral.
5	N_perc	0.25	Below the 30% inverted U peak — novelty below this level is insufficient.

5	P	0.60	Some knowledge expected and positive.
5	Q_rel	0.50	Zero knowledge is already problematic.
5	R	0.50	Stranger to stranger is genuine baseline. Truly symmetric.
5	AC	0.35	Optimal at 0.5 by design of $4R(1-R)$. Neutral is genuinely 0.5.
6	L	0.75	Low consequence already below neutral — anything touching your world registers.
6	M	0.50	High consistency with established knowledge is strongly expected.
			Physical evidence is rare. Its absence is neutral; presence is a strong boost.

10.1 Magnitude Normalization for Multiplicative Aggregation

Multiplicative aggregation of variables in $[0,1]$ produces systematic magnitude compression: the product of n variables each in $[0,1]$ converges toward zero as n grows, regardless of individual values. This is a well-known property of multiplicative aggregation, not a modeling error. Clusters 3, 4, and 5 contain more multiplicative terms than Clusters 1 and 2, producing composite neutral points well below the expected $[0.35, 0.45]$ range. The standard correction is a monotonic root transformation applied to the composite output before it enters the likelihood ratio function. Root transformations are standard in multiplicative index normalization and preserve the strict ordering of scores while restoring the composite neutral to its intended range. The formulas and individual $x0_c$ values are unchanged.

Cred_final = Cred(s) [no transform] $x0 = 0.301$

Stmt_final = Stmt(s) [no transform] $x0 = 0.354$

Deliv_final = $\sqrt{\text{Deliv}(s,r)}$ [1/2 root] $x0 = 0.379$

Sit_final = $\sqrt{\text{Sit}(s,r)}$ [1/2 root] $x0 = 0.436$

Recv_final = $\sqrt[3]{\text{Recv}(s,r)}$ [1/3 root] $x0 = 0.450$

The cube root applied to Recv is justified by the greater number of compounding multiplicative terms in that cluster relative to Deliv and Sit. The square root applied to Deliv and Sit reflects moderate compression requiring moderate correction. λ_c for Clusters 3, 4, and 5 now uses the post-transformation $x0_c$ values: $x0_Deliv = 0.379$, $x0_Sit = 0.436$, $x0_Recv = 0.450$. Clusters 1 and 2 are unaffected.

11. Limit Definitions and Singularity Handling

Condition	Behavior	Resolution
B_eff approaching 0	Cred approaches 0	Limit definition: $\lim(B \rightarrow 0) B^{(B^{-0.5})} = 0$ for B in $[0,1]$

X approaching 0	Deliv approaches 0	Limit definition: statement to oneself has no delivery score
H approaching 0	Stmnt approaches 0	Analytic: $\lim(H \rightarrow 0) (1 - e^{(-kH)}) = 0$
R = 0 or 1	Recv approaches 0	Analytic: $4R(1-R) = 0$ at both extremes
AC = 0	Recv = 0	Analytic: $0^{1.15} = 0$
AB = 0	Sit = 0	Analytic: $0^{2.1} = 0$
P approaching 0	Recv volatile	Open Problem 4 — requires careful limit analysis

12. Propositions and Proof Sketches

Proposition 1 — Strict Superset Hierarchy

Statement: The input space of $T_abs(s)$ is a proper subset of the input space of $T_perc(s,r)$, which is a proper subset of the input space of $I(s,r)$. Formally: $I_{abs} \subset I_{perc} \subset I_I$. The scores T_abs , T_perc , and I are real-valued functions defined on these nested spaces; the subset relationship holds between the domains, not the scalar outputs.

Proof sketch: Follows directly from Axiom 5 by construction. I_abs contains the input variables of Clusters 1, 2, and 6 only. I_perc adds the input variables of Clusters 3, 4, and 5 — all receiver-dependent. I_I adds importance exponents and substitutes N_imp for N_perc in Recv. No input of I_abs is absent from I_perc or I_I ; no input of I_perc is absent from I_I . The converse fails by counterexample — AB (opposing voices) is an element of I_perc and I_I but not I_abs . Therefore $I_{abs} \subset I_{perc} \subset I_I$. QED by Axiom 5.

Proposition 2 — Convergence of Perception to Truth

Statement: As P approaches 1 and all perceptual inputs approach their neutral points, $T_perc(s,r)$ approaches $T_abs(s)$.

Proof sketch: T_perc is formed by updating T_abs on Deliv, Sit, Recv sequentially. For T_perc to equal T_abs , all three likelihood ratios must equal 1 — each cluster must hit its neutral composite score. Under the neutral state N^* (all variables at their $x0_c$ values), every $\Lambda_c = 1$ by definition of $x0_c$. The rolling prior is unchanged. Therefore posterior odds equal prior odds, and $T_perc = T_abs$. The $P \rightarrow 1$ condition is necessary because $N_perc(1-P) \rightarrow N_perc^0 = 1$ as $P \rightarrow 1$, eliminating novelty's contribution and completing the collapse of Recv to its neutral point. Formal monotonicity proof deferred — see Proposition 6, Open Problem.

Proposition 3 — Importance Strict Dominance

Statement: The input space I_I of $I(s,r)$ strictly contains the input space I_perc of $T_perc(s,r)$, which strictly contains the input space I_abs of $T_abs(s)$. As scoring functions, I , T_perc , and T_abs are defined on strictly nested domains.

Proof sketch: The input space of I contains all inputs of T_perc plus the importance exponents and N_imp substitution. These additional structural elements cannot be recovered from T_perc alone — setting importance exponents to 1 and $N_imp = N_perc$ recovers a projection-like restriction only after adding structural information not present in T_perc 's output. Therefore the input spaces are strictly nested by construction. Axioms 4, 5, 11 jointly establish the hierarchy. The stronger claim that truth scores are literal projections of the importance function requires a precise definition of scoring space as a measure space and is deferred to the projection open problem.

Proposition 4 — Importance Receiver Necessity

Statement: There exists no well-defined $I(s)$ without a specified receiver r .

Proof: $I(s,r)$ depends on $\text{Recv}(s,r)$ which contains AC, Q_rel , R, P — all undefined without a receiver r . By Axiom 4, importance without a receiver is undefined. Since $\text{Recv}(s,r)$ is undefined without r , and $I(s,r)$ depends on $\text{Recv}(s,r)$, $I(s,r)$ is undefined without r . QED.

Proposition 5 — Boundedness of Importance

Statement: For every admissible statement s and receiver r , the revised importance score satisfies $I(s, r) \in [0, 1]$.

Proof: By construction, $I_{\text{context}}(s, r) \in [0, 1]$. For the non-degenerate case $I_{\text{context}} \in (0, 1)$, the context odds $O_{\text{context}} = I_{\text{context}} / (1 - I_{\text{context}})$ are non-negative and finite. The likelihood ratio $\Lambda_{I,T}(T_{\text{abs}})$ is strictly positive for all $T_{\text{abs}} \in [0, 1]$. Therefore $O_I = \Lambda_{I,T} O_{\text{context}}$ is non-negative and finite, and the logistic recovery $I = O_I / (1 + O_I)$ lies strictly between 0 and 1. In the boundary cases, $I_{\text{context}} = 0$ gives $O_I = 0$ and therefore $I = 0$; $I_{\text{context}} = 1$ is handled by the standard odds-limit, giving $I = 1$. Therefore $I(s, r) \in [0, 1]$ by construction. QED.

12.5 Limitations and Scope

The framework should be read as a formal architecture and research program, not as a fully empirically calibrated scoring instrument. The core structural claims are mathematical and conceptual: receiver-independent truth evidence, perceived truth, and importance use nested but non-identical input spaces; receiver-independent truth must not be contaminated by delivery and personal consequence; and importance must not be collapsed into truth. The numerical parameters are deliberately exposed rather than hidden.

Several limitations follow directly from this status. First, many input variables still require operational measurement functions. Q_rel , AA, AC, AB, and the domain relevance weights $w(s,i)$ are semantically clear but not yet measurement-complete. Second, the neutral point table is a set of explicit semantic judgments. It is better to declare these judgments than to bury them inside free constants, but they remain open to empirical recalibration. Third, the Bayesian update engine assumes conditional independence of cluster evidence given A. That assumption is tractable and transparent, but real evidence streams often contain covariance, shared upstream sources, echo chambers, and strategic coordination. Fourth, T_abs is written as $T_abs(s)$ for simplicity, but several stress tests show that evaluation time matters; a fully formal implementation should use $T_abs(s,t)$. Fifth, the stress tests are semantic audits, not validation against ground-truth datasets. They show that the architecture behaves correctly on deliberately hard edge cases; they do not prove empirical accuracy.

These limitations do not weaken the central model; they locate the work precisely. The present paper fixes the architecture, exposes the parameters, states the open problems, and prevents the most obvious formal failure modes. Empirical calibration, covariance modeling, and population-level aggregation are the next layer rather than hidden assumptions in the current layer.

13. Open Problems

#	Problem	Status
1	Variable operationalization: formal measurement functions for Q_rel , AA, and remaining intuition-based inputs.	Open

2	Likelihood functions: P(cluster_score A) and P(cluster_score not-A) for each update. Partially resolved via $\text{Lambda}(x) = \exp((x-x_0)/(x_0(1-x_0)))$. K_c per cluster requires definition.	Partially resolved
3	Projection proof: formally proving that T_{abs} and T_{perc} are obtainable as restrictions of I to I_{abs} and I_{perc} respectively — i.e. that the scalar truth scores can be recovered as projections of the importance function onto their respective input subspaces. The input-space hierarchy $I_{\text{abs}} \subset I_{\text{perc}} \subset I_I$ is established by construction (Proposition 1); the stronger projection claim is the remaining unproven part.	Open
4	P approaching 0 singularity: behavior of $\text{Recv}(s,r)$ as receiver prior knowledge approaches zero.	Open
5	Corroboration decay: formally defining α in $\text{Lambda}_n = 1 + (\text{Lambda}_{\text{base}} - 1)/n^\alpha$.	Open
6	Steepness constant k : derived as $k = \ln(100) = 4.605$ from $\text{max}=0.99$ condition. Locked.	Resolved
7	Composite neutral points: RESOLVED via root transformations applied to cluster outputs before likelihood ratio translation. $\text{Cred}^*=0.301$ (no transform), $\text{Stmt}^*=0.354$ (no transform), $\text{Deliv}^*=\sqrt{0.144}=0.379$, $\text{Sit}^*=\sqrt{0.190}=0.436$, $\text{Recv}^*=\sqrt[3]{0.091}=0.450$. Transformations are monotone — score ordering preserved, only magnitude corrected. Lambda_c operates on post-transformation scores with updated x_{0_c} values: $x_{0_Deliv}=0.379$, $x_{0_Sit}=0.436$, $x_{0_Recv}=0.450$. Cube root applied to Recv justified by greater compounding of multiplicative terms relative to Deliv and Sit .	Resolved

8	Weight derivation $w(s,i)$: formal computation of domain relevance weights from statement subject analysis.	Open
9	Q_rel operationalization: formal measurement of receiver-speaker relationship strength.	Open
10	Window size w : empirical justification for default $w=10$ in rolling prior.	Open
11	Proposition 6 (Knowledge Monotonicity): formally prove T_perc monotonically converges toward T_abs as P increases.	Open
12	Empirical validation: testing framework outputs against ground truth datasets of statement credibility and importance ratings.	Open
13	N normalization: validate $N_norm = 1 - e^{(-N)}$ as appropriate normalization for KL divergence inputs.	Open

14. Declared Empirical Parameters

The following parameters are acknowledged as empirical rather than axiomatically derived. They represent reasonable defaults pending empirical calibration:

- $w = 10$: rolling prior window size.
- $x0_c$ table values: all neutral points in Section 10 are semantic judgments, not derived values. Subject to empirical revision.
- N_perc exponents 0.7 and 1.5: derived from 30% peak threshold, which itself is an empirical estimate of where novelty begins to hurt perceived truth.
- α in Λ_n : decay rate of corroboration likelihood ratios. Undefined pending empirical study of echo chamber dynamics.
- $\beta = 0.15$: dampening factor for the T_abs likelihood ratio in the I engine. Controls how strongly receiver-independent truth evidence updates importance. $\beta=1$ would give undampened Bayesian update (as in T_perc). $\beta=0$ would make T_abs invisible to the I engine. The value 0.15 was chosen so that near-zero T_abs produces a mild negative Λ (approximately 0.75) rather than a catastrophic penalty, while near-unity T_abs produces a mild positive Λ (approximately 1.35) rather than an overwhelming boost. This preserves correct ranking across all ten stress-test cases while formally instantiating Axiom 11. Subject to empirical recalibration.
- M context-dependence floor: $x0_M$ varies between 0.40 and 0.50 depending on statement type. Boundaries of this variation require domain-specific calibration.

14.1 New Open Problems Identified via Stress Testing

- Open Problem 14 — Temporal dependence of T_abs : T_abs is formulated as time-independent but evidence quality assessments (particularly M and K) are time-dependent. The Powell WMD

case showed $T_{abs}=0.606$ in February 2003 (evidence appeared credible) versus near-zero post-2005 (evidence known fabricated). A formal treatment requires temporal indexing $T_{abs}(s,t)$ where t is the evaluation time.

- Open Problem 15 — Population-level importance $I_{pop}(s)$: $I(s,r)$ is strictly receiver-relative by Axiom 4, but the importance of statements like political misinformation is not about one receiver — it is about millions of receivers with wildly different T_{perc} values. A formal definition of $I_{pop}(s)$ as an integral of $I(s,r)$ over a receiver distribution would capture societal importance while preserving the receiver-relative foundation.

14.2 Empirical Stress Tests

Ten statements were evaluated using the full framework to verify semantic correctness, correct ranking behaviour, and robustness across the four critical edge-case types: (a) false but consequential statements, (b) true but unimportant statements, (c) prestigious sources in wrong domains, and (d) unfalsifiable claims. All variable assignments reflect epistemic state at time of statement, not hindsight knowledge. The $\beta=0.15$ dampened Bayesian update architecture is used throughout for I .

Statement T_{abs} T_{perc} I

1	Powell WMD (UN, Feb 2003)	0.606	0.990	0.995
2	Trump inflation claim (2025-26)	0.000175	0.00149	0.937
3	COVID vaccine tweet (Oct 2022)	0.000828	0.00433	0.798
4	Boring Truth (Earth revolves Sun)	0.99996	0.99990	0.301
5	Climate expert flood warning	0.99992	0.99999	0.993
6	Whistleblower (corp. fraud)	0.688	0.910	0.904
7	Prestigious Lie (Pauling/cancer)	0.403	0.631	0.783
8	5G COVID claim (viral, 2020)	0.000355	0.000675	0.524
9	Unfalsifiable claim (platitude)	0.009	0.0097	0.355
10	True Rumour (anon. acq. tip)	0.405	0.626	0.886

Semantic audit and ranking verification. All rankings are semantically correct. Three ranking triplets hold cleanly: (1) Powell (0.995) > Trump (0.937) > COVID tweet (0.798), capturing the gradient from a Secretary of State at the UN to a presidential rally claim to an anonymous social media post; (2) Climate expert (0.993) > Whistleblower (0.904) > Pauling lie (0.783), capturing the gradient from a verified expert warning to directly affected parties, through a documented but unverified allegation, to a sincere but domain-displaced Nobel laureate; (3) True Rumour (0.886) > 5G claim (0.524) > Unfalsifiable platitude (0.355), capturing the gradient from market-moving insider information through viral conspiracy to a private consolatory platitude.

Critical edge cases passed. (a) False but consequential: Trump ($T_{abs}=0.000175$, $I=0.937$) and COVID tweet ($T_{abs}=0.000828$, $I=0.798$) score high importance because the dampened Λ (0.747 and 0.754 respectively) allows massive context to dominate. Under the prior direct-multiplication architecture ($I = I_{context} \times T_{abs}^{0.25}$), these scored 0.110 and 0.143 — correctly diagnosed as wrong. (b) True but unimportant: Boring Truth ($T_{abs}=0.99996$, $I=0.301$) correctly scores low because $T_{abs}^{0.5}$ near unity gives a mild $\Lambda=1.350$, and the context (zero stakes, zero engagement, zero novelty) drives $I_{context}=0.242$. Truth alone cannot inflate importance — Axiom 11 is satisfied. (c) Prestigious source in wrong domain: Pauling ($T_{abs}=0.403$, $I=0.783$) correctly scores moderate importance. The Nobel name inflates the receiver’s perceived truth ($T_{perc}=0.631$) and the desperate receiver’s $AC=0.99$ drives high context. But T_{abs} correctly reflects that wrong-domain expertise produces only moderate objective truth-probability. (d) Unfalsifiable claim: $H=0.02$ collapses Stmt to 0.014, which cascades to $T_{abs}=0.009$

and $I=0.355$ — low but non-zero because the friend relationship and emotional context give the platitude moderate receiver-relative importance. Axiom 8 (falsifiability as truth condition) is correctly instantiated.

Architecture validation. The $\beta=0.15$ dampened update replaces the prior direct-multiplication formula $I = I_{context} \times T_{abs}^{0.25}$. The old formula failed two cases catastrophically: Trump scored 0.110 ($T_{abs}^{0.25}$ penalty crushes a consequential false claim) and Boring Truth scored 0.242 with no differentiation between high-truth and low-context. The new formula correctly solves both: Beta dampens the Lambda of near-zero T_{abs} from 0.143 to 0.747 (allowing consequence to dominate for false claims), and dampens the Lambda of near-unity T_{abs} from 7.39 to 1.35 (preventing truth alone from inflating unimportant statements). The dampening factor $\beta=0.15$ formally instantiates Axiom 11 by ensuring T_{abs} carries strictly less evidential weight in I than in T_{perc} throughout.

15. Conclusion

This paper presents a unified mathematical framework for truth-probability and importance-scoring of statements, grounded in 14 formal axioms, six structured input clusters, a derived likelihood ratio function with cluster-specific neutral points, and sequential Bayesian updating throughout. The central contributions are: the three-output architecture with formally established input-space hierarchy; the separation of receiver-independent truth evidence from perceived truth with T_{abs} as anchor for T_{perc} ; the treatment of importance as the full scoring space of which truth scores are modeled as conjectural projections; the derivation of the falsifiability steepness constant k from a maximum condition; the identification and patching of a topological vulnerability in naive consistency-falsifiability coupling; and the dampened Bayesian update architecture for $I(s,r)$ — separating $I_{context}$ from truth-weighting via a dampening factor $\beta=0.15$ that formally instantiates Axiom 11, prevents false but consequential statements from being penalised, and prevents true but inconsequential statements from being inflated.

The framework is both theoretically novel in its unified architecture and practically tractable in its input structure. Fifteen open problems are precisely identified, providing a clear agenda for formalization, empirical grounding, and eventual publication.

Appendix A: Notation Reference

Symbol	Meaning
$T_{abs}(s)$	Receiver-independent evidential truth posterior of statement s
$T_{perc}(s,r)$	Perceived truth-probability of s for receiver r
$I(s,r)$	Importance score of s for receiver r
$P(A)$	Prior probability that statement s is objectively true
$\Lambda_c(x)$	Likelihood ratio for cluster c at score x
$x0_c$	Neutral point of cluster c — score at which $\Lambda_c = 1$
$B_{eff}(s)$	Statement-weighted effective bias scalar
$C_{eff}(s)$	Statement-weighted effective expertise scalar
$w(s,i)$	Domain relevance weight of bias/expertise dimension i for statement s
Z_{eff}	Bias-adjusted speaker stakes
N	Raw KL divergence: $KL(P_s Q_r)$
N_{norm}	Normalized novelty: $1 - e^{(-N)}$
N_{perc}	Novelty for T_{perc} (inverted U): $4.18 * N_{norm}^{0.7} * (1 - N_{norm})^{1.5}$
N_{imp}	Novelty for $I(s,r)$ (monotonic): $1 - e^{(-N(1+N))}$
k	Falsifiability steepness constant: $\ln(100) = 4.605$
w	Rolling prior window size: default 10

alpha	Corroboration decay exponent (undefined — Open Problem 5)
beta	Dampening factor for T_abs likelihood ratio in I engine. Default value 0.15. Controls evidential weight of objective truth on importance; beta=1 gives undampened Bayesian update, beta=0 makes T_abs invisible to I. Formally instantiates Axiom 11 (beta < 1 ensures T_abs exerts strictly less evidential weight on I than on T_perc).

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